

ORDERING INFORMATION

Device	Temperature Range	Package
MC1747F	-55°C to +125°C	Ceramic Flat
MC1747G	-55°C to +125°C	Metal Can
MC1747L	-55°C to +125°C	Ceramic DIP
MC1747CF	0°C to +75°C	Ceramic Flat
MC1747CG	0°C to +75°C	Metal Can
MC1747CL	0°C to +75°C	Ceramic DIP
MC1747CP2	0°C to +75°C	Plastic DIP

MC1747 MC1747C

DUAL MC1741 INTERNALLY COMPENSATED, HIGH PERFORMANCE OPERATIONAL AMPLIFIER

... designed for use as summing amplifiers, integrators, or amplifiers with operating characteristics as a function of the external feedback components. The MC1747L and MC1747CL are functionally, electrically, and pin-for-pin equivalent to the μ A747 and μ A747C respectively.

- No Frequency Compensation Required
- Short-Circuit Protection
- Wide Common-Mode and Differential Voltage Ranges
- Low-Power Consumption
- No Latch Up
- Offset Voltage Null Capability

FIGURE 1 — HIGH-IMPEDANCE, HIGH-GAIN
INVERTING AMPLIFIER

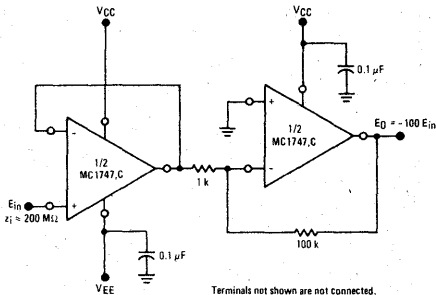
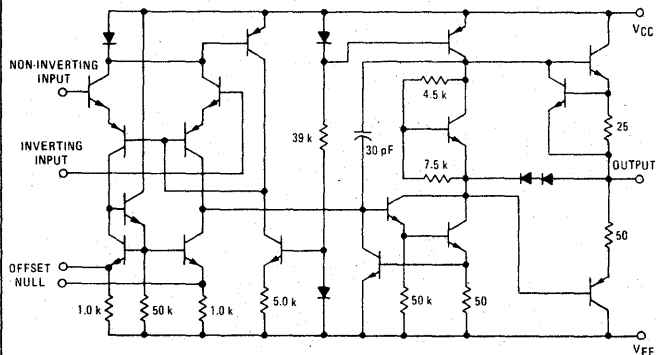


FIGURE 2 — CIRCUIT SCHEMATIC



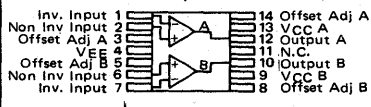
Circuit diagrams utilizing Motorola products are included as a means of illustrating typical semiconductor applications; consequently, complete information sufficient for construction purposes is not necessarily given. The information has been carefully checked and is believed to be entirely reliable. However, no responsibility is assumed for inaccuracies. Furthermore, such information does not convey to the purchaser of the semiconductor devices described any license under the patent right of Motorola Inc. or others.

(DUAL MC1741)

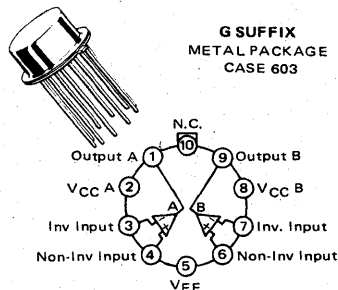
DUAL OPERATIONAL AMPLIFIER

SILICON MONOLITHIC
INTEGRATED CIRCUIT

F SUFFIX
CERAMIC PACKAGE
CASE 607

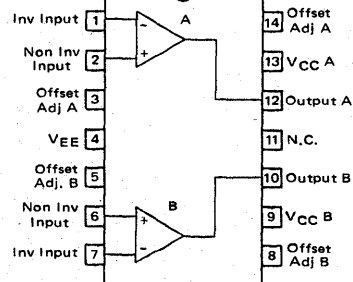


G SUFFIX
METAL PACKAGE
CASE 603



P2 SUFFIX
PLASTIC PACKAGE
CASE 646

L SUFFIX
CERAMIC PACKAGE
CASE 632-02
TO-116



MC1747, MC1747C

MAXIMUM RATINGS (T_A = +25°C unless otherwise noted.)

Rating	Symbol	MC1747	MC1747C	Unit
Power Supply Voltages	V _{CC} V _{EE}	+22 -22	+18 -18	Vdc
Differential Input Signal Voltage ①	V _{ID}	± 30		Volts
Common-Mode Input Swing Voltage ②	V _{ICR}	± 15		Volts
Output Short-Circuit Duration	t _{OS}	Continuous		
Voltage (Measurement between Offset Null and V _{EE})		± 0.5		Volts
Operating Ambient Temperature Range	T _A	-55 to +125	0 to +75	°C
Storage Temperature Range	T _{stg}	-65 to +150	-65 to +150	°C
Junction Temperature Ceramic and Metal Package Plastic Package	T _J	175 150		°C

ELECTRICAL CHARACTERISTICS (V_{CC} = +15 Vdc, V_{EE} = -15 Vdc, T_A = +25°C unless otherwise noted.)

Characteristics	Symbol	MC1747			MC1747C			Unit
		Min	Typ	Max	Min	Typ	Max	
Input Bias Current T _A = +25°C T _A = T _{high} ③ T _A = T _{low} ③	I _{IB}	—	80 30 300	500 500 1500	—	80 30 30	500 800 800	nAdc
Input Offset Current T _A = +25°C T _A = T _{high} T _A = T _{low}	I _{IO}	—	20 7.0 85	200 200 500	—	20 7.0 7.0	200 300 300	nAdc
Input Offset Voltage (R _S ≤ 10 kΩ) T _A = +25°C T _A = T _{low} to T _{high}	V _{IO}	—	1.0 1.0 6.0	5.0 6.0 —	—	1.0 1.0 —	6.0 7.5 —	mVdc
Offset Voltage Adjustment Range		—	± 15	—	—	± 15	—	mV
Differential Input Impedance (Open-loop, f = 20 Hz) Parallel Input Resistance Parallel Input Capacitance	r _i C _i	0.3 —	2.0 1.4	— —	0.3 —	2.0 1.4	— —	MΩ pF
Common-Mode Input Voltage Swing T _{low} ≤ T _A ≤ T _{high}	V _{ICR}	± 12	± 13	—	± 12	± 13	—	Volts
Common-Mode Rejection Ratio (R _S = 10 kΩ) T _{low} ≤ T _A ≤ T _{high}	CMRR	70	90	—	70	90	—	dB
Open-Loop Voltage Gain T _A = +25°C T _A = T _{low} to T _{high} } (V _O = ±10 V, R _L = 2.0 kΩ)	A _{vol}	50,000 25,000	200,000 —	— —	25,000 15,000	200,000 —	— —	Volts
Transient Response (Unity Gain) (V _{in} = 20 mV, R _L = 2.0 kΩ, C _L ≤ 100 pF) Rise Time Overshoot Percentage	t _{PLH}	— —	0.3 5.0	— —	— —	0.3 5.0	— —	μs %
Slew Rate (Unity Gain)	SR	—	0.5	—	—	0.5	—	V/μs
Output Impedance	z _o	—	75	—	—	75	—	ohms
Short-Circuit Output Current	I _{OS}	—	25	—	—	25	—	mAdc
Channel Separation		—	120	—	—	120	—	dB
Output Voltage Swing (T _{low} ≤ T _A ≤ T _{high}) R _L = 10 kΩ R _L = 2.0 kΩ	V _{OR}	± 12 ± 10	± 14 ± 13	— —	± 12 ± 10	± 14 ± 13	— —	V _{pk}
Power Supply Sensitivity (T _{low} to T _{high}) V _{EE} = Constant, R _S ≤ 10 kΩ V _{CC} = Constant, R _S ≤ 10 kΩ	PSS+ PSS-	— —	30 30	150 150	— —	30 30	150 150	μV/V
Power Supply Current (each amplifier) T _A = +25°C T _A = T _{low} T _A = T _{high}	I _{CC} , I _{EE}	— — —	1.7 2.0 1.5	2.8 3.3 2.5	— — —	1.7 2.0 2.0	2.8 3.3 3.3	mAdc
DC Power Consumption (each amplifier) T _A = +25°C T _A = T _{low} T _A = T _{high}	P _C	— — —	50 60 45	85 100 75	— — —	50 60 60	85 100 100	mW

① For supply voltages of less than ± 15 V, the maximum differential input voltage is equal to ± (V_{CC} + |V_{EE}|).

② For supply voltages of less than ± 15 V, the maximum input voltage is equal to the supply voltage (±V_{CC} - |V_{EE}|).

③ T_{low}: 0°C for MC1747CL

-55°C for MC1747L

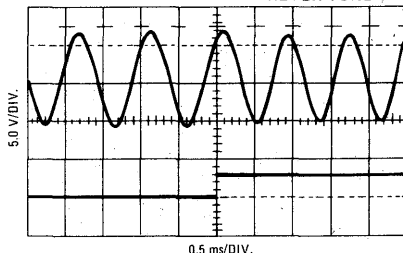
T_{high}: +75°C for MC1747CL

+125°C for MC1747L



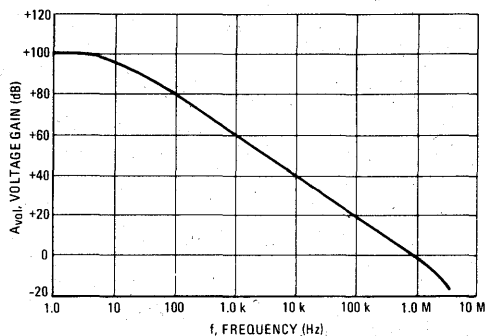
MOTOROLA Semiconductor Products Inc.

FIGURE 4 – TYPICAL FREQUENCY-SHIFT KEYSER TONE GENERATOR

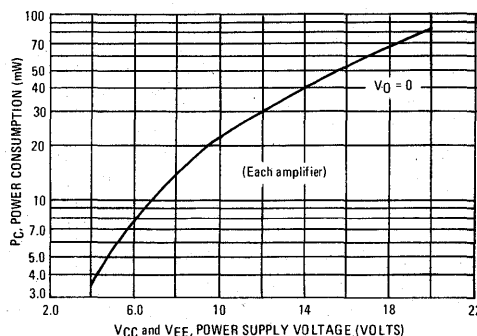


($V_{CC} = +15$ Vdc, $V_{EE} = -15$ Vdc, $T_A = +25^{\circ}\text{C}$ unless otherwise noted.)

FIGURE 6 – OPEN-LOOP FREQUENCY RESPONSE



**FIGURE 8 – POWER CONSUMPTION
versus POWER SUPPLY VOLTAGE**



TYPICAL CHARACTERISTICS (continued)

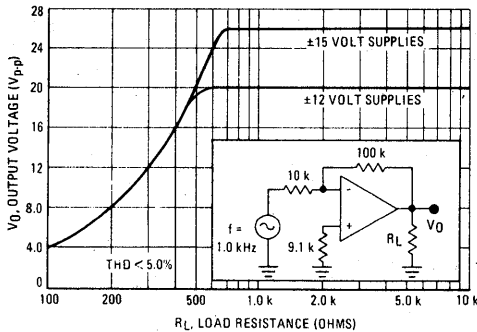
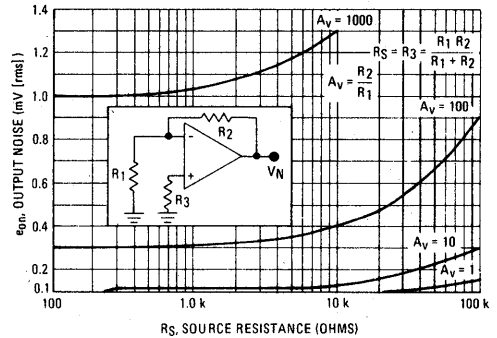
(V_{CC} = +15 Vdc, V_{EE} = -15 Vdc, T_A = +25°C unless otherwise noted.)FIGURE 9 — OUTPUT VOLTAGE SWING
versus LOAD RESISTANCE

FIGURE 10 — OUTPUT NOISE versus SOURCE RESISTANCE



THERMAL INFORMATION

The maximum power consumption an integrated circuit can tolerate at a given operating ambient temperature, can be found from the equation:

$$P_D(T_A) = \frac{T_J(\max) - T_A}{R_{\theta JA}(\text{Typ})}$$

Where: $P_D(T_A)$ = Power Dissipation allowable at a given operating ambient temperature. This must be greater than

the sum of the products of the supply voltages and supply currents at the worst case operating condition.

$T_J(\max)$ = Maximum Operating Junction Temperature as listed in the Maximum Ratings Section

T_A = Maximum Desired Operating Ambient Temperature

$R_{\theta JA}(\text{Typ})$ = Typical Thermal Resistance Junction to Ambient

